

PHASE 4 — PROJECT DOCUMENTATION

Hospital Appointment Management System

Project Title: Hospital Appointment Management System

Platform: ServiceNow (Scoped Application)

Application Scope: x_2065601_hospit_0

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1. Project Overview

The **Hospital Appointment Management System** is a ServiceNow scoped application developed to simplify and manage the process of scheduling hospital appointments.

Traditional appointment systems often depend on phone calls, manual registers, or disconnected processes. These approaches can result in double-bookings, difficulty in checking doctor availability, missed appointments, and inefficient handling of cancelled appointment slots.

The proposed system provides a centralized ServiceNow application where patients can register and book appointments based on available doctor slots. Doctors can manage their availability and daily patient queue, while front-desk or administrative users can manage appointments and perform manual bookings.

The system also includes a waitlist mechanism that allows waiting patients to be promoted automatically when an appointment slot becomes available.

2. Problem Statement

Hospitals and clinics may face difficulties when appointment scheduling is handled manually or through fragmented systems.

The major problems identified are:

- Difficulty in checking real-time doctor availability.
- Possibility of assigning the same slot to multiple patients.
- Manual handling of appointment cancellations.

- No systematic waitlist management.
- Difficulty for doctors to manage their daily patient queue.
- Limited visibility of appointment information.
- Additional workload for front-desk staff.

These problems create delays for patients and increase the administrative effort required to manage appointments.

3. Proposed Solution

The Hospital Appointment Management System provides a centralized ServiceNow-based solution for managing patients, doctors, availability slots, appointments, and waitlisted patients.

The application uses a **ServiceNow Scoped Application** with a custom Service Portal.

The major workflow is:

Patient Registration → Doctor Selection → Slot Availability → Appointment Booking → Check-In → Consultation → Completion

If an appointment is cancelled or a slot becomes available, the system can identify a suitable waitlisted patient and promote that patient to the available slot.

The application also uses role-based access so that patients, doctors, and front-desk/admin users see and perform actions appropriate to their responsibilities.

4. Project Scope

4.1 In-Scope Features

The following features are included in the project:

- Patient registration.
- Doctor registration.
- Doctor specialization and qualification management.
- Doctor availability management.
- Appointment slot management.
- Patient appointment booking.

- Appointment history.
- Appointment cancellation.
- Appointment status management.
- Doctor daily queue management.
- Patient check-in.
- Appointment completion.
- No-show handling.
- Waitlist management.
- Automatic waitlist promotion.
- Front-desk manual booking.
- Role-based access control.
- Service Portal interface.
- Backend verification using ServiceNow tables.

4.2 Out-of-Scope Features

The following features are outside the current project scope:

- Hospital billing and payment processing.
- Electronic Health Record integration.
- Prescription management.
- Insurance verification.
- Telemedicine integration.
- Mobile application.
- Advanced SMS/email notification services.

These features can be considered as future enhancements.

5. Technology Used

Technology	Purpose
ServiceNow	Application platform
Scoped Application	Application isolation and development
Service Portal	User-facing portal
AngularJS	Service Portal widget development
HTML	Portal interface

CSS	User interface styling
JavaScript	Client-side and server-side logic
GlideRecord	Database record operations
ACL	Role-based security
ServiceNow Tables	Application data storage

6. Application Scope

The application was developed as a ServiceNow scoped application.

Application Scope: `x_2065601_hospit_0`

The scoped application provides separation of application components and allows custom tables, fields, scripts, portal components, and security configurations to be managed within the project.

7. System Architecture

The application consists of four major layers:

7.1 User Interface Layer

The Service Portal provides the interface for different users.

The main portal widget is:

`hospital_booking`

The portal contains sections for:

- Home
- Patient Registration
- Patient Booking & History
- Doctor Registration
- Doctor Queue & Slots

- Front Desk/Admin
- Waitlist

7.2 Application Logic Layer

The application logic processes operations such as:

- Checking available slots.
- Booking appointments.
- Cancelling appointments.
- Updating appointment states.
- Managing doctor queues.
- Handling waitlisted patients.

7.3 Data Layer

ServiceNow custom tables store patient, doctor, availability, appointment, and waitlist information.

7.4 Security Layer

Custom ACLs and role-based flows control access to application functions and records.

8. Data Model

8.1 Patient Table

Table: `x_2065601_hospit_0_patient`

Field	Purpose
patient_id	Unique patient identifier
patient_name	Patient name
age	Patient age
blood_group	Patient blood group
phone	Contact number

city	Patient city
address	Patient address
primary_issue	Primary medical issue

8.2 Doctor Table

Table: x_2065601_hospit_0_doctor

Field	Purpose
doctor_id	Unique doctor identifier
doctor	Reference to ServiceNow user
specialization	Doctor specialization
qualification	Doctor qualification
max_patients_per_day	Maximum daily patient capacity
shift	Doctor working shift
consultation_fee	Consultation fee
slot_time	Default slot duration

8.3 Doctor Availability Slot Table

Table: x_2065601_hospit_0_doctor_availability_slot

Field	Purpose
doctor	Reference to doctor
slot_date	Date of availability
start_time	Slot starting time
end_time	Slot ending time

status Available / Booked / Cancelled / Blocked

8.4 Appointment Table

Table: x_2065601_hospit_0_appointment

Field	Purpose
appointment_id	Unique appointment identifier
patient	Reference to patient
doctor	Reference to doctor
slot	Reference to availability slot
state	Booked / Checked-In / Completed / Cancelled / No-Show
reason_for_visit	Patient's reason for appointment

8.5 Waitlist Table

Table: x_2065601_hospit_0_waitlist

Field	Purpose
patient	Reference to patient
doctor	Reference to doctor
preferred_date	Patient's preferred date
priority_order	Position in waitlist
status	Waiting / Promoted / Expired / Cancelled

9. Service Portal

The Service Portal acts as the primary user interface for the application.

9.1 Home

The Home section provides an entry point to the hospital appointment system and directs users to the relevant functionality.

9.2 Patient Registration

Patients can enter their required information and create their patient record.

The registration process stores the submitted information in the Patient table.

9.3 Patient Booking & History

Patients can:

1. Select a doctor.
2. Check available appointment slots.
3. Select a suitable slot.
4. Provide the reason for the visit.
5. Book the appointment.
6. View existing appointments.
7. Cancel an appointment when required.

9.4 Doctor Registration

The Doctor Registration section allows authorized users to create and manage doctor information, including specialization, qualification, shift, and daily capacity.

9.5 Doctor Queue & Slots

Doctors can manage their availability and view their daily patient queue.

The queue supports appointment lifecycle activities such as:

- Check-in.
- Consultation completion.
- No-show marking.

9.6 Front Desk/Admin

Front-desk or administrative users can manage appointment-related activities and perform manual bookings for walk-in patients.

9.7 Waitlist

The Waitlist section allows patients who could not obtain their preferred appointment slot to be placed in a waiting queue.

When an eligible slot becomes available, the system can promote a waitlisted patient.

10. Appointment Booking Process

The appointment booking process follows these general steps:

Step 1 — Patient Registration

The patient enters the required personal information.

Step 2 — Doctor Selection

The patient selects a doctor based on the required specialization.

Step 3 — Availability Check

The application checks the doctor's availability slots.

Step 4 — Slot Selection

The patient selects an available date and time.

Step 5 — Booking

The selected slot is reserved and the appointment record is created.

Step 6 — Appointment Confirmation

The appointment is stored in the Appointment table with the appropriate state.

11. Double-Booking Prevention

One of the important technical requirements of the project is preventing two users from booking the same appointment slot.

The application implements **atomic slot booking**.

The basic process is:

Check Slot → Lock/Update Slot Status → Create Appointment

The availability slot changes from:

Available → Booked

and the appointment record is created as part of the booking transaction.

This prevents the same available slot from being assigned to multiple patients.

12. Appointment Lifecycle

The appointment lifecycle uses defined states.

Booked → Checked-In → Completed

Alternative states include:

Booked → Cancelled

and

Booked → No-Show

The lifecycle allows doctors and front-desk staff to track the current status of every appointment.

13. Waitlist Automation

The system provides an automated process for handling cancelled or freed appointment slots.

For example:

1. A patient books a doctor slot.
2. Another patient is placed on the waitlist for the same doctor/date.
3. The booked appointment is cancelled.
4. The slot becomes available.
5. The system checks the waitlist.
6. An eligible waiting patient is promoted.
7. The waitlist status changes from **Waiting** to **Promoted**.

This reduces the need for manual intervention.

14. Role-Based Access Control

Different users have different responsibilities within the application.

User Role	Main Responsibilities
Patient	Registration, booking, history, cancellation
Doctor	Availability, patient queue, check-in/completion/no-show
Front Desk/Admin	Registration support, manual booking, appointment management

ACLs are used to restrict unauthorized access to application data and operations.

15. Backend Verification

A major feature of the application is that portal operations are connected to actual ServiceNow backend records.

When users perform actions through the portal, corresponding records are created or updated in the ServiceNow tables.

For example:

Patient Portal → Patient Table

Appointment Booking → Appointment + Availability Slot Tables

Waitlist → Waitlist Table

This allows administrators and developers to verify the application through ServiceNow list views and backend records.

16. Testing Summary

The application should be validated using functional, security, and concurrency testing.

Important scenarios include:

- Patient registration.
- Doctor registration.
- Slot creation.
- Appointment booking.
- Duplicate booking prevention.
- Appointment cancellation.
- Waitlist promotion.
- No-show handling.
- Role-based access.
- Front-desk booking.
- Empty-state handling.

Testing confirms that the major application workflows operate according to the defined requirements.

17. Challenges Faced and Solutions

Challenge 1 — Preventing Double Booking

Multiple users may attempt to select the same available slot.

Solution: Atomic slot-booking logic was used so that the slot status is changed and the appointment is created as part of the booking operation.

Challenge 2 — Connecting Portal Fields with Backend Fields

Incorrect internal field names can cause portal submissions to fail or update the wrong fields.

Solution: Portal field mappings were aligned with the internal names of the ServiceNow tables.

Challenge 3 — Managing Cancelled Slots

Manually identifying patients waiting for cancelled slots can increase administrative work.

Solution: A waitlist automation process was designed to identify eligible waiting patients and promote them when a slot becomes available.

18. Screenshots

ServiceNow Scoped Application Overview

The screenshot displays the ServiceNow Studio interface for a Scoped Application named "Hospital Appointment Management". The interface is divided into several sections:

- Left Sidebar:** Contains a navigation menu with categories like Automation, Client Development, Data, Properties, Security, Server Development, User Interface, and Other. The "Data" category is currently selected.
- Top Bar:** Shows the application name "Hospital Appointment Management" and a "Publish" button.
- Main Content Area:** Displays a table of application components. The table has columns for Name, Type, Updated, Updated by, and Created by. The components are organized into sections: Data, User Interface, and Automation.

Name	Type	Updated	Updated by	Created by
Data				
Waitlist	Table	2026-08-16 23:...	System Administr...	System Administr...
Appointment	Table	2026-08-16 23:...	System Administr...	System Administr...
Patient	Table	2026-08-16 23:...	System Administr...	System Administr...
Doctor Availability Slot	Table	2026-08-16 23:...	System Administr...	System Administr...
Doctor	Table	2026-08-16 23:...	System Administr...	System Administr...
User Interface				
No user interface yet. Start on the front-end by adding an interface, like a workspace or portal.				
Automation				
An automation user. Automation streamlines processes and workflows within your organization.				

Patient Registration Form

The screenshot shows the 'Patient - Create PAT0002008' form in ServiceNow. The form is titled 'Patient New record' and includes a 'Submit' button. The form fields are organized into two columns. The left column contains: Patient ID (text input with value PAT0002008), Patient Name (text input), Age (text input), Blood Group (dropdown menu with value -- None --), and Phone (text input). The right column contains: City (text input), Address (text input), and Primary Issue (text input). The form is displayed on a light blue background with a white border. The ServiceNow logo and navigation bar are visible at the top.

Patient ID	PAT0002008	City	
Patient Name		Address	
Age		Primary Issue	
Blood Group	-- None --		
Phone			

Submit

Doctor Registration Form

The screenshot shows the 'Doctor - Create DOCT0001009' form in ServiceNow. The form is titled 'Doctor New record' and includes a 'Submit' button. The form fields are organized into two columns. The left column contains: Doctor ID (text input with value DOCT0001009), Doctor (text input with a search icon), Specialization (dropdown menu with value -- None --), and Qualification (text input). The right column contains: Max Patients Per Day (text input), Shift (dropdown menu with value -- None --), Consultation Fee (text input), and Slot Time (text input). The form is displayed on a light blue background with a white border. The ServiceNow logo and navigation bar are visible at the top.

Doctor ID	DOCT0001009	Max Patients Per Day	
Doctor		Shift	-- None --
Specialization	-- None --	Consultation Fee	
Qualification		Slot Time	

Submit

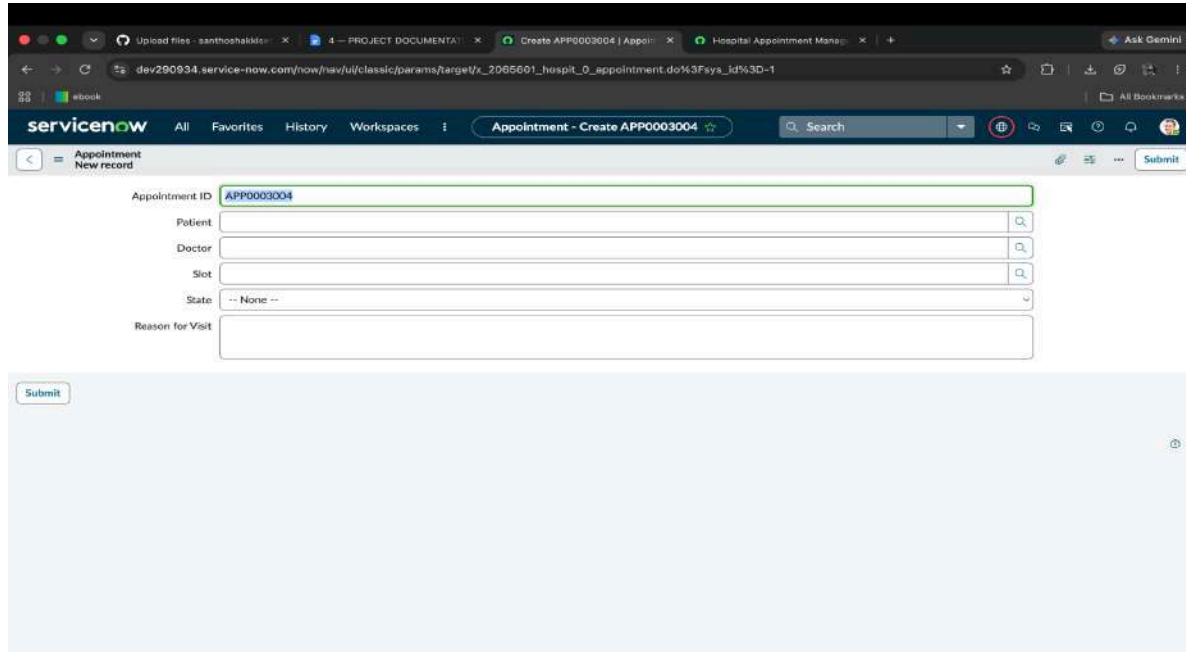
Doctor Availability Slot Form

The screenshot shows a web browser window with the ServiceNow interface. The browser tabs include 'Upload Files - saitheshak...', 'PROJECT DOCUMENTAT...', 'Create Created | Doctor Avail...', and 'Hospital Appointment Mana...'. The address bar shows the URL: 'dev290934.service-now.com/now/nav/ui/classic/params/target/x_2065601_hospit_0_doctor_availability_slot.do%3Fsys_id%3D0-1'. The ServiceNow header includes the logo, navigation links (All, Favorites, History, Admin), a search bar, and a 'Doctor Availability Slot - Create Create...' button. The main content area is titled 'Doctor Availability Slot' and 'New record'. It contains a form with the following fields: 'Doctor' (text input), 'Start Time' (calendar icon), 'End Time' (calendar icon), 'Status' (dropdown menu with '-- None --'), and 'Slot Date' (calendar icon). A 'Submit' button is located at the bottom left of the form area.

Patient Appointment Booking Page

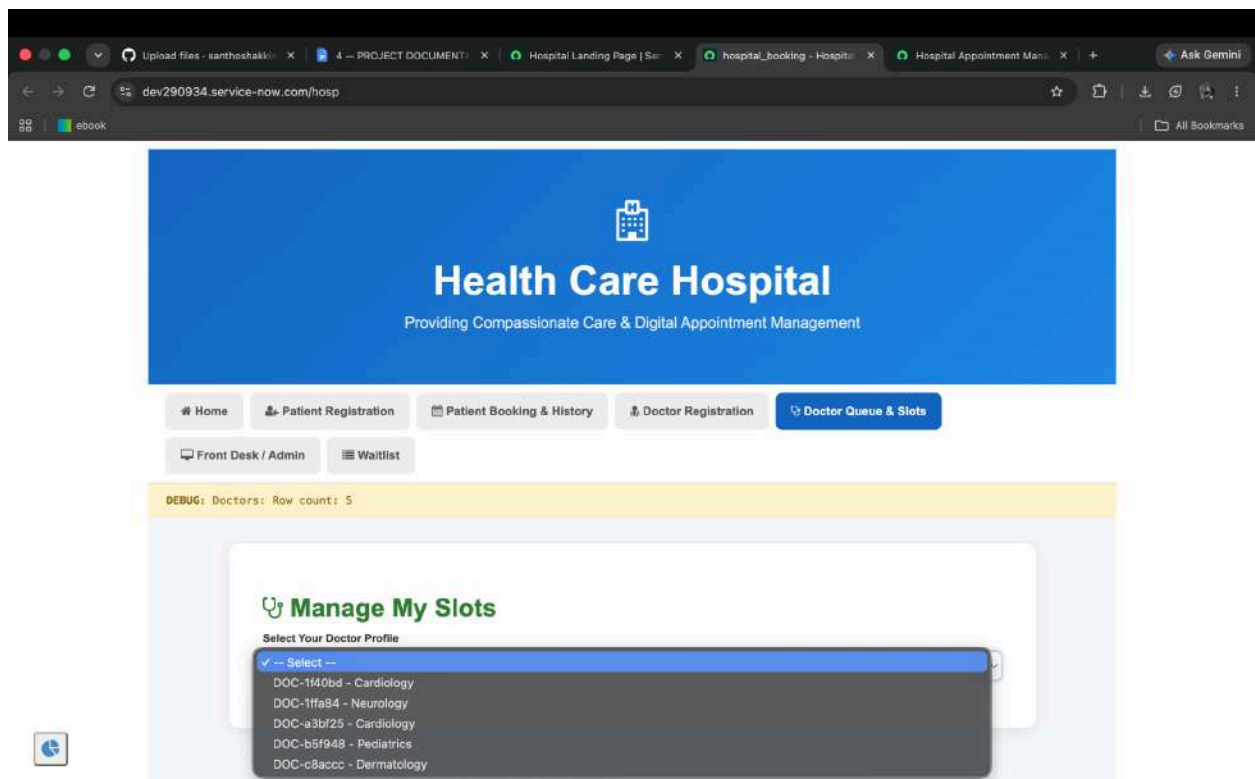
The screenshot shows a web browser window with the ServiceNow interface. The browser tabs include 'ServiceNow Developers', 'Doctor Availability Slots | Ser...', 'hospital_booking - Hospital I...', and 'My Name is Billa AK Vid...'. The address bar shows the URL: 'dev290934.service-now.com/hosp'. The ServiceNow header includes the logo, navigation links (All, Favorites, History, Admin), a search bar, and a 'Hospital Appointment Mana...' button. The main content area is titled 'Health Care Hospital' and 'Providing Compassionate Care & Digital Appointment Management'. It features a navigation bar with links: 'Home', 'Patient Registration', 'Patient Booking & History', 'Doctor Registration', 'Doctor Queue & Slots', 'Front Desk / Admin', and 'Waitlist'. Below the navigation bar, there are two main sections: 'Patient Portal' and 'Doctor & Staff Portal'. The 'Patient Portal' section includes a wheelchair icon, the text 'Patient Portal', a description 'Register profile, search doctor slots, book appointments, and track status.', and a blue button labeled 'Login / Register as Patient'. The 'Doctor & Staff Portal' section includes a doctor icon, the text 'Doctor & Staff Portal', a description 'Manage availability slots, view daily patient queue.', and a green button labeled 'Login / Register as Doctor'.

Appointment Form in ServiceNow



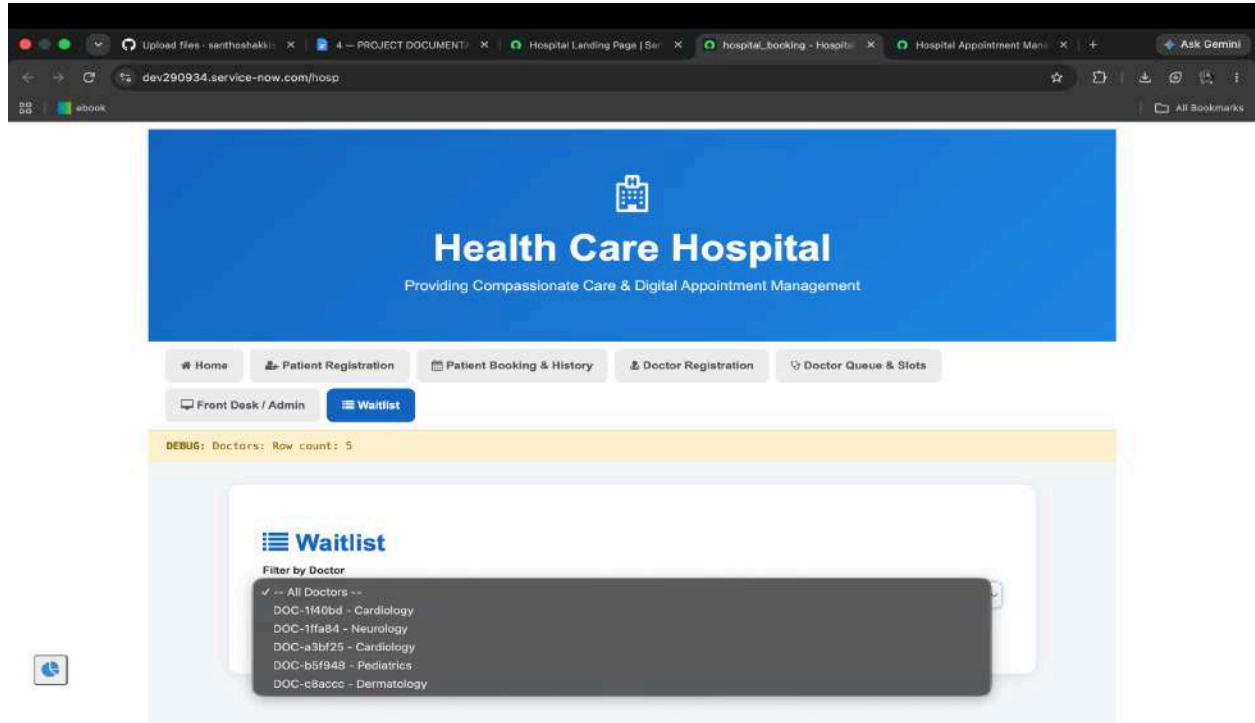
The screenshot shows a web browser window with the ServiceNow interface. The browser tabs include "Upload files - santhoshakki...", "PROJECT DOCUMENT...", "Create APP0003004 | Appoi...", and "Hospital Appointment Man...". The address bar shows the URL "dev290934.service-now.com/now/hav/ui/classic/params/target?x_2065601_hospit_0_appointment.do%3Fsys_id%3D-1". The ServiceNow header includes the logo, navigation tabs (All, Favorites, History, Workspaces), and a search bar. The main content area is titled "Appointment - Create APP0003004" and contains a form with the following fields: "Appointment ID" (pre-filled with "APP0003004"), "Patient", "Doctor", "Slot", "State" (dropdown menu with "-- None --"), and "Reason for Visit". A "Submit" button is located at the bottom right of the form.

Doctor Queue & Slots Page

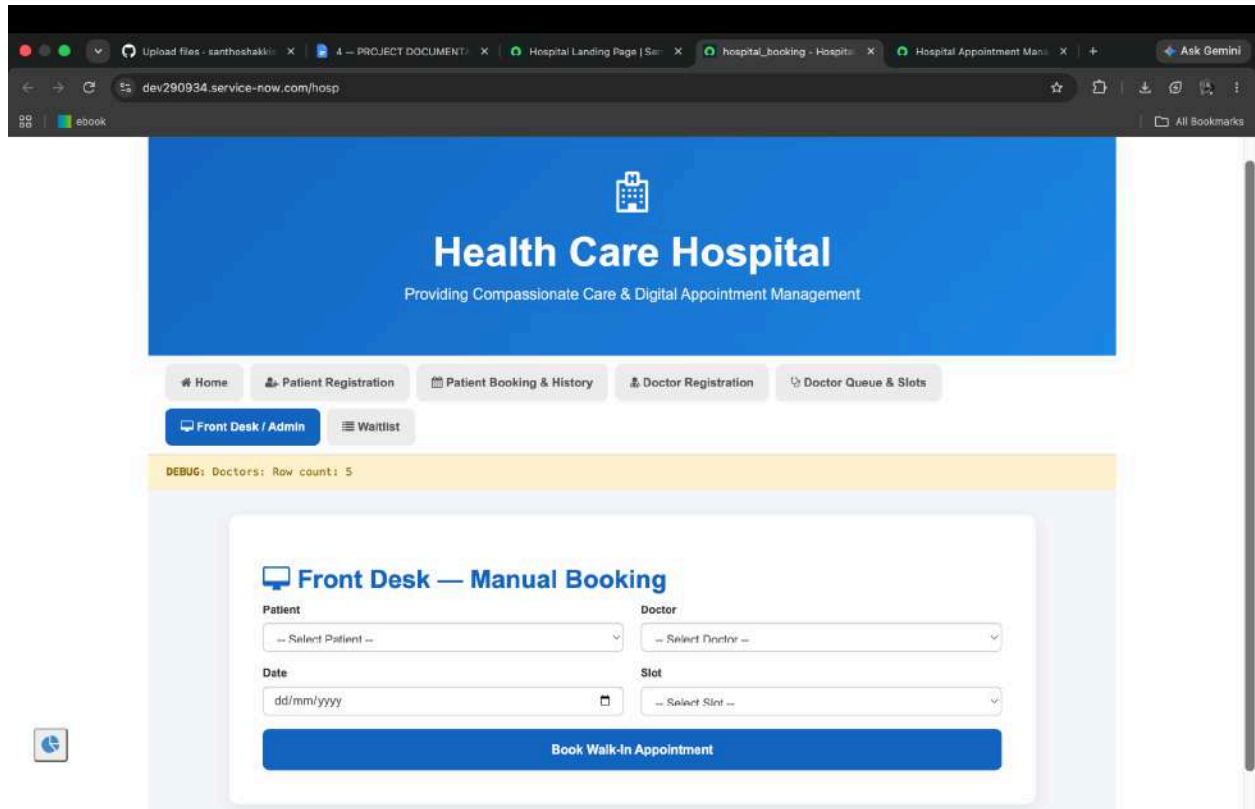


The screenshot shows a web browser window with the ServiceNow interface. The browser tabs include "Upload files - santhoshakki...", "PROJECT DOCUMENT...", "Hospital Landing Page | S...", "hospital_booking - Hospiti...", and "Hospital Appointment Man...". The address bar shows the URL "dev290934.service-now.com/hosp". The ServiceNow header includes the logo, navigation tabs (All, Favorites, History, Workspaces), and a search bar. The main content area is titled "Health Care Hospital" and features a blue header with the text "Providing Compassionate Care & Digital Appointment Management". Below the header is a navigation bar with buttons for "Home", "Patient Registration", "Patient Booking & History", "Doctor Registration", and "Doctor Queue & Slots" (highlighted). Below the navigation bar is a "Front Desk / Admin" button and a "Waitlist" button. A yellow banner displays the text "DEBUG: Doctors: Row count: 5". The main content area is titled "Manage My Slots" and contains a "Select Your Doctor Profile" dropdown menu. The dropdown menu is open, showing a list of doctor profiles: "DOC-1f40bd - Cardiology", "DOC-1ffa84 - Neurology", "DOC-a3bf25 - Cardiology", "DOC-b5f948 - Pediatrics", and "DOC-c8accc - Dermatology".

Waitlist Management Page



Front Desk/Admin Booking Page



19. Future Enhancements

The current application can be extended with additional healthcare-related functionality.

Potential enhancements include:

- SMS appointment reminders.
- Email notifications.
- Telemedicine integration.
- Insurance verification.
- Online payment processing.
- Mobile application.
- Integration with Electronic Health Records.
- Advanced hospital analytics and dashboards.
- Doctor performance reports.
- Patient notification and reminder workflows.
- These features can be considered for future versions of the application.

20. Conclusion

The **Hospital Appointment Management System** demonstrates how ServiceNow can be used to develop a structured healthcare appointment management solution using a scoped application. The system centralizes patient registration, doctor management, availability slots, appointment booking, appointment lifecycle management, and waitlist handling.

The use of Service Portal provides a user-friendly interface, while ServiceNow tables provide structured backend data storage. GlideRecord-based application logic supports backend operations, and ACLs provide role-based security. The implementation of atomic slot booking addresses the problem of double-booking, while waitlist automation improves the handling of cancelled appointment slots.

Overall, the project demonstrates practical application of ServiceNow application development concepts including **Scoped Applications, Custom Tables, Service Portal, AngularJS Widgets, GlideRecord, ACLs, and workflow/automation concepts** in a real-world hospital appointment management scenario.

21. Tools and References

Platform

- ServiceNow Scoped Application

Development Components

- Service Portal
- AngularJS
- HTML
- CSS
- JavaScript
- GlideRecord
- ACLs
- ServiceNow Custom Tables

Application Scope `x_2065601_hospit_0`

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